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Research on surface defect classification model of automotive flange disks based on machine vision

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Abstract

To address the inefficiency and instability of manual classification for surface defects in automotive flange disks, this paper proposes an integrated SVM classification framework combining improved Canny edge detection with enhanced Hu moment features. The dataset construction process involves capturing 50 raw images of each category, including sand holes, scratches, and defect-free samples, using an industrial camera. A 4× data augmentation strategy was applied, including mirror flipping, random panning within ±10 pixels, and gamma correction with span coefficients of 0.5 to 1.5 to generate a standardized dataset of 600 images. These were partitioned into 420 training samples and 180 independent testing samples. During preprocessing, the improved bilateral filtering (gradient similarity instead of grayscale similarity) improves PSNR by 12.4% compared to traditional Gaussian filtering. Extended Sobel operators incorporating 45° and 135° gradient templates improved annular workpiece feature representation, integrated with Otsu-based adaptive dual-threshold segmentation. For feature extraction, a six-dimensional modified Hu moment descriptor was developed through normalized second-order central moments, constructing scale-invariant ρ1-ρ6 feature vectors that eliminate discrete-scale sensitivity. Experimental validation via five-fold cross-validation using 480 training samples per fold optimized SVM parameters, implementing a third-order polynomial kernel with a penalty factor C = 76.8. The model demonstrated 97.2%±0.6% classification accuracy on independent test sets, surpassing baseline SVM by 13.5% while maintaining efficient inference speed of 0.684 s per image. Comparative analysis shows that the performance is better than MobileNetV3, with an increase in accuracy and recall of 0.7% and 1.8%, respectively. Furthermore, its classification accuracy is comparable to that of EfficientNetB0 (97.4%), while its inference speed is significantly faster—approximately 9.1 times. This optimized framework provides methodological advancements and practical insights for industrial quality inspection applications.

Keywords Flange disk · Defect classification · Improved canny · Enhanced hu moments · SVM

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1 Introduction

With the rapid development of China's economy and the continuous improvement in living standards, the rapid growth of the automotive industry has placed significant demands on vehicle safety performance. The flange disk, one of the crucial components of a vehicle, ensures a secure attachment of the wheel hub to the axle. The quality of the flange disk directly determines the overall safety performance of the vehicle. Currently, the two main detection methods used in the production process of automotive flange disk are visual inspection and the machine vision inspection [1]. However, with the improvement in productivity and the urgent need for intelligent development, visual inspection is highly susceptible to errors due to fatigue or other subjective factors, leading to missed or false inspections. Therefore, visual inspection is no longer adequate to meet the high precision and efficiency requirements in the production inspection process. More and more enterprises have adopted machine vision to replace human inspection in assessing the shape, dimensions, and surface quality of flange disks [2].

Deep learning techniques in machine vision have significantly improved detection efficiency and accuracy. However, there are two major challenges when applying deep learning in practical scenarios. On the one hand, its outstanding performance heavily relies on a large volume of labeled data [3] and high computational resource consumption [4]. On the other hand, in scenarios with limited computational resources, the high demand for computing power [5] makes the creation of large-scale and diverse defect datasets highly costly. This process is not only time-consuming and labor-intensive but also risks causing unnecessary losses in production.

In summary, when applied to defect detection [6], it is crucial to maintain a balance between cost, resources, and performance. Scholars both domestically and internationally have conducted extensive research on this topic. For instance, Li Yawen [7] et al. explored the optimization of the SVM classification model using genetic algorithms, combined with Hu moment feature extraction. Liu Ruiyuan [8] et al. proposed a machine vision-based method for detecting surface defects in automotive precision parts, utilizing Hu moments and SVM to achieve automatic detection and classification of part surface defects. Zhu Jiajun [9] et al. introduced a PSO-SVM-based method for detecting surface defects on steel strips, successfully improving SVM performance by 12%, enabling fast and accurate identification and classification of steel strip surface defects. Zhang Zhiwen [10] et al. proposed a method for improving emotion recognition accuracy by fusing three physiological signals: electroencephalography (EEG), electromyography (EMG), and electrodermal activity (EDA). Mao Yuezhong [11] et al. developed a custom rapid photoionization detector-gas chromatography-electronic nose system for quality grading. The features derived from detection signals were input into support vector machines and principal component analysis, demonstrating good performance in classifying 132 samples into three different grades. Wu Miao [12] et al. proposed a deep transfer learning-based CAD system, using VGG19 to extract deep features from ultrasound images. The classification, conducted by k-nearest neighbors (KNN) and support vector machine (SVM) classifiers, achieved an overall classification accuracy of 92.01 ± 1.48%. Tan Lizhi [13] et al. developed a machine vision-based approach for automotive spark plug defect detection by integrating an improved Histogram of Oriented Gradients (HOG) feature extraction method with a Support Vector Machine (SVM) classifier. Experimental results demonstrated a detection accuracy of 94%. Liang Jianwei [14] et al. proposed a machine vision-based framework for detecting surface defects on automotive lamp covers, employing an improved Mean Shift algorithm combined with Fuzzy C-Means (FCM) clustering. The method achieved superior performance, with average precision (95.25% and 94.96%) and recall rates (93.13% and 93.03%) in both single-image and multi-image scenarios. Xie Dan [15] et al. introduced a machine vision algorithm incorporating morphological image processing, Gabor filtering, and gamma grayscale enhancement for detecting typical defects such as scratches, abrasions, and spots on windshield wipers. The system achieved an average detection speed of 0.724 s per wiper with a precision of 95%. Qi Jinlong [16] et al. proposed an improved VGG16-based deep learning model for automotive component defect detection by optimizing the fully connected layer architecture and integrating an Angular Margin Fisher-Softmax (AMF-Softmax) loss function. The improved model achieved a test accuracy of 95.26%. Cai Bingbin [17] et al. designed a lightweight BDDNet-based machine vision system for real-time surface defect inspection of automotive bearings. The system achieved an average detection time of 0.735 s per bearing on a CPU platform, with a detection accuracy of 90.02%.

Based on the aforementioned research—Support Vector Machines (SVM) is a powerful supervised learning algorithm. Additionally, machine learning-based defect detection methodologies exhibit superior operational efficiency by directly extracting discriminative defect features. These approaches demonstrate significantly lower computational complexity and improved real-time performance. Therefore, this paper employs an SVM classification model to analyze the dataset obtained from the experiments. The flange dataset employed in this paper exhibits homogeneous background characteristics with distinctly discernible defect features, which facilitates robust feature extraction and precise defect localization. Under these conditions, a classification method combining the improved Canny edge detection with Hu moment feature extraction and SVM training is proposed. The main contributions of this paper are as follows:

To further improve the classification performance of flange disk defect detection in the field of industrial automation, this paper presents a comprehensive optimization of the traditional Canny edge detection algorithm. The specific optimization strategies and their advantages are as follows: (1) In the denoising phase, the improved bilateral filtering algorithm (using gradient similarity instead of grayscale similarity and spatial distance weight fusion) algorithm is used to replace the Gaussian filter, which effectively suppresses the image noise while retaining the edge information. (2) In the gradient calculation step, the Sobel operator is extended by adding detection templates at 45° and 135° to enrich the extraction of edge information. (3) In the threshold determination process, the Otsu algorithm is introduced to automate the adaptive determination of double thresholds. (4) After completing the edge detection, the enhanced Hu moment shape features are further extracted, thereby providing more stable feature support for subsequent defect classification. Finally, through a comparison of the performance of four kernel functions, the polynomial kernel function is selected to train the SVM, resulting in the improved CannyHu-SVM algorithm.

The structure of this paper is organized as follows: Sect. 2 presents the theoretical foundations of the algorithm and the specific improvements made; Sect. 3 provides the experimental results and analysis; finally, Sect. 4 offers a comprehensive conclusion of the entire paper.

2 Principles and improvements of Canny, Hu moments and SVM

2.1 Improved Canny edge detection algorithm

The standard Canny algorithm has several limitations: Gaussian filtering may disrupt the continuity of edges; gradient calculation using a 2 × 2 neighborhood is prone to interference from noise, which can lead to the loss of edge information; and the manually selected double thresholds lack adaptability, making it difficult to handle diverse image environments [18]. To address these issues, the following improvements are proposed:

(1) Improved Bilateral Filtering Replaces Gaussian Filtering

Bilateral filtering is an advanced nonlinear technique [19] that simultaneously considers both spatial proximity and grayscale similarity. Then, the combined weights of the two are used as convolution kernels to perform convolution on the noisy images to achieve denoising.

The bilateral filtering calculation is given by Eq. (1):

$$I(x,y) = \frac{1}{W_p}\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{i,j}(x,y)f(x+i,y+j) \tag{1}$$

In the equation, the filtered pixel values $I(x, y)$ is a weighted average of the pixel value $f(x+i, y+j)$ in the N×N window around (x, y) of the original image, with the weight $\omega_{i,j}(x, y)$ taking into account both spatial proximity and grayscale similarity, and the normalization factor W_P (sum of all weights) ensuring that the pixel value falls between 0 and 255.

In order to solve the problem that traditional bilateral filtering may lead to image distortion or false edges when protecting edges, a solution was proposed: replacing the gray similarity with gradient similarity, and constructing a new bilateral filtering method based on spatial distance.

The gradient bilateral filtering calculation is given by Eq. (2):

$$\begin{cases} I^{,}(x,y) = \dfrac{1}{W_p}\displaystyle\sum_{i=1}^{m}\sum_{j=1}^{n}\omega_{i,j}g_{\sigma_s}(d_p,q)\times \\ g_{\sigma_r}(r_p,q)\left[I(x+i,y+j)-I(x,y)\right] \\ \omega_{i,j} = \exp\dfrac{\left\{-\left[(i-i^{,})^2+(j-j^{,})^2\right]\right\}}{2\sigma_s^2}\times \\ \exp\dfrac{\left\{-\left[f(i,j)-f(i^{,},j^{,})\right]^2\right\}}{2\sigma_r^2} \end{cases} \tag{2}$$

In the equation, the output pixel value $I^{,}(x, y)$ is calculated from a filter window centered on (x, y) and size $(m \times n)$: for each pixel $(i^{,},j^{,})$ in the neighborhood, its weight $\omega_{i,j}$ is determined by the spatial distance weight $g_{\sigma s}(d_p,q)$ and the gray difference weight $g_{\sigma r}(r_p,q)$, where (d_p,q) is the spatial distance between $(i^{,},j^{,})$ and the center point, and (r_p,q) is the difference between the two gray values f$(i^{,},j^{,})$ and $f(i, j)$. The final result is derived from a weighted average.

The core difference between gradient bilateral filtering and standard bilateral filtering is that weight $\omega_{i,j}$ incorporates gradient information: the larger the gradient between adjacent pixels (p, q) and (x, y), the higher the weight value, thus enhancing the edge response; The smaller the gradient, the lower the weight value, which in turn strengthens the noise suppression. This mechanism allows it to optimize both edge retention and noise reduction.

(2) Sobel Gradient Calculation with Additional 45° and 135° Directions

The gradient magnitude is a key indicator of pixel intensity variation in an image. By extending the traditional Sobel operator in the horizontal and vertical directions to include the 45° and 135° directions, the first-order partial derivatives are computed for a 8-pixel neighborhood in four directions. The template calculation is given by Eq. (3):

$$\left\{\begin{array}{l} H_{\mathrm{x}}=\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} \\ H_{45^\circ}=\begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} \end{array}\right., \left\{\begin{array}{l} H_{\mathrm{y}}=\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \\ H_{135^\circ}=\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -2 \end{bmatrix} \end{array}\right. \quad (3)$$

The convolution templates for the gradients in the four directions are denoted by H_x,H_y,$H_{45°}$ and $H_{135°}$. The gradient calculation is given by Eq. (4):

$$\left\{\begin{array}{l} d_{\mathrm{x}}(i,j)=l(i+1,j)-l(i-1,j) \\ d_{\mathrm{y}}(i,j)=l(i,j+1)-l(i,j-1) \\ d_{45^\circ}(i,j)=l(i-1,j+1)-l(i+1,j-1) \\ d_{135^\circ}(i,j)=l(i+1,j+1)-l(i-1,j-1) \end{array}\right. \quad (4)$$

$$\left\{\begin{array}{l} f_x(i,j)=d_x(i,j)+\frac{d_{45^\circ}(i,j)+d_{135^\circ}(i,j)}{2} \\ f_y(i,j)=d_y(i,j)+\frac{d_{45^\circ}(i,j)-d_{135^\circ}(i,j)}{2} \end{array}\right. \quad (5)$$

$$\left\{\begin{array}{l} M(i,j)=\sqrt{\alpha\cdot f_x(i,j)^2+\beta\cdot f_y(i,j)^2} \\ M=\sqrt{M_x^2+M_y^2+M_{45^\circ}^2+M_{135^\circ}^2} \end{array}\right. \quad (6)$$

Equation (4) represents the partial derivatives of the gradient template in the four directions, while in Eq. (5), $f_x(i, j)$ and $f_y(i,j)$ represent the gradients in the horizontal and vertical directions, respectively. In Eq. (6), the gradient weight coefficients are denoted by α and β, with the gradient components in each direction represented by M_x,M_y,$M_{45°}$ and $M_{135°}$. The final gradient M can then be obtained.

(3) Non-Maximum Suppression

When determining edge pixels, the direction of the gradient must also be considered. For instance, if the gradient intensity in the horizontal direction (x-direction) exceeds that in the vertical direction (y-direction), the decision of whether a pixel belongs to an edge will primarily rely on the gradient in the horizontal direction. Conversely, if the gradient in the vertical direction is more significant, the decision will emphasize the vertical gradient. By incorporating the consideration of gradient direction, the accuracy of edge detection is enhanced.

(4) Otsu's Thresholding Segmentation

After non-maximum suppression, erroneous edge data often remains in the image [20]. To eliminate these effects, this paper employs Otsu's double thresholding. The Otsu segmentation algorithm automates the determination of optimal double thresholds, subjective judgments.

The specific calculation formula is as follows: Let there be a threshold t, which divides the image into two classes: background (denoted as $C0$) and foreground (denoted as $C1$). The probabilities of a pixel being classified into $C0$ and $C1$ are $p0$ and $p1$, respectively. The mean grayscale values of $C0$ and $C1$ are denoted by $m0$ and $m1$, respectively, and the global average grayscale value of the image is m. The calculation is given by Eq. (7):

$$\begin{cases} p0\times m0+p1\times m1=m \\ p0+p1=1 \\ \sigma^2=p0(m0-m)^2+p1(m1-m)^2 \end{cases} \quad (7)$$

In the equation, σ^2 is maximized, and the optimal threshold t is selected as the high threshold for double thresholding, while the low threshold is set to half of the high threshold.

2.2 Enhancment of Hu moment features

The Hu moments, proposed by M.K. Hu from the University of Hong Kong [21], are a set of seven invariant moments used for image feature description. These invariant moments efficiently capture the global shape features of an image and possess scale and rotational invariance, along with low computational complexity. Additionally, Hu moments demonstrate strong robustness to noise, making them highly suitable for the work scenario in this paper. However, when performing scaling operations on a discrete image, the normalized central moments also change, as described in Eq. (8). Additionally, the relationship between the seven invariant moments under the influence of the scale factor σ is given by Eq. (9):

$$\eta'_{pq}=\frac{\mu'_{pq}}{\mu_{00}'^{\left(1+\frac{p+q}{2}\right)}}=\frac{\sigma^{p+q}\mu_{pq}}{\mu_{00}^{\left(1+\frac{p+q}{2}\right)}}=\sigma^{p+q}\eta_{pq} \quad (8)$$

$$\begin{cases} h'_1=\sigma^2h_1 \\ h'_2=\sigma^4h_2 \\ h'_3=\sigma^6h_3 \end{cases}, \begin{cases} h'_4=\sigma^6h_4 \\ h'_5=\sigma^{12}h_5 \\ h'_6=\sigma^8h_6 \\ h'_7=\sigma^{12}h_7 \end{cases} \quad (9)$$

As shown in the above equations, the value of the normalized central moments depends on the order $(p+q)$ and is influenced by the scale factor σ. To maintain the invariance of the Hu moments with respect to the scale factor in the discrete state, this paper improves the Hu moments using $h_{2'}$(computed based on normalized central moments) as a reference and constructs a new set of invariant moments, as given in Eq. (10):

$$\begin{cases} \rho_1 = \dfrac{(h_1')^2}{h_2'} \\ \rho_2 = \dfrac{(h_3')^2}{(h_2')^3} \\ \rho_3 = \dfrac{(h_4')^2}{(h_2')^3} \end{cases}, \begin{cases} \rho_4 = \dfrac{h_5'}{(h_2')^3} \\ \rho_5 = \dfrac{h_6'}{(h_2')^2} \\ \rho_6 = \dfrac{h_7'}{(h_2')^3} \end{cases} \tag{10}$$

In the equation, ρ represents the value of h' after eliminating the scaling factor σ. Based on the above equation, in the discrete state, these six feature values can achieve invariance to scaling, while also maintaining invariance to rotation and translation.

Specific operations: (1) first calculate the geometric moment and central moment of the image, (2) normalize the central moment of the image, (3) calculate the seven standard Hu moments, (4) analyze the influence of the scale factor on the normalized central moment under discrete scaling, and (5) take $h2'$ as the benchmark, adjust the other moments to eliminate the scale factor, and construct a new six invariant moments ρ.

Table 1 shows the comparison of the enhanced Hu moments with HOG and SIFT methods in the key indicators of feature extraction. The results show that the enhanced Hu moment shows comprehensive performance advantages: in terms of data dependence, its unsupervised characteristics only need < 100 samples are significantly better than those of HOG requires 200 samples and SIFT requires 500 samples. In terms of computing efficiency, the real-time processing of Hu moments < 10ms CPU is 50% faster than HOG (15-20ms) and 5–10 times faster than SIFT (50-100ms), which is due to its six-dimensional lightweight design; In terms of deployment cost, the memory embedded adaptability of Hu moments is only 22% of HOG and 18% of SIFT, and its mathematical transparency reduces the difficulty of debugging, and can be directly deployed on the hardware platform shown in Fig. 2. In terms of noise robustness, the Hu moments maintain a > 90% recognition rate in the interference environment, which is significantly higher than that of HOG and SIFT, because the global characteristics of the shape moment are insensitive to local disturbances, while the local descriptors of HOG/SIFT are easily affected by dust interference.

2.3 Principle and advantages of support vector machine (SVM)

The fundamental principle of SVM is to separate data points in a multi-dimensional space with the goal of finding an optimal hyperplane that best distinguishes between different classes of data [22]. In the case where the data is perfectly linearly separable, the objective is to find a hyperplane that maximizes the margin between the two classes, as illustrated in Fig. 1.

SVM also demonstrates significant advantages in classification tasks, particularly in terms of generalization capability, often outperforming decision trees, random forests, and K-Nearest Neighbors (K-NN). In high-dimensional spaces, SVM can effectively address non-linear problems using the kernel trick, a task that decision trees struggle to handle directly. Compared to K-NN, SVM offers higher computational efficiency during the prediction phase, is more robust to noise, and is less prone to overfitting. In contrast to random forests, the SVM model is more streamlined and tends to perform better on small-scale datasets, also exhibiting less susceptibility to overfitting. Overall, SVM demonstrates greater stability when dealing with features of varying scales, maintaining classification performance while minimizing computational resource consumption. Table 2 outlines the advantages and disadvantages of commonly used traditional machine learning methods [23].

2.3.1 Feature input

The enhanced Hu moments are input to the SVM classifier in a fixed six-dimensional sequence [ρ_1, ρ_2, ρ_3, ρ_4, ρ_5, ρ_6] after standardization: (1) standardization: Z-score normalization is applied to eliminate magnitude discrepancies. (2) vector assembly: construct the feature vector as [ρ_1, ρ_2, ρ_3, ρ_4, ρ_5, ρ_6]. (3) kernel mapping: the vector is mapped via a 3rd-order polynomial kernel (d = 3) to a 1536-dimensional space for classification.

Table 1 Comparison between enhanced Hu moments and traditional methods

Dimension	Enhanced Hu moments	HOG	SIFT
Data dependency	Unsupervised (< 100)	Medium(200+)	High(500+)
Computational cost	< 10ms (CPU)	15-20ms	50-100ms
Interpretability	Mathematically transparent	Medium	Medium
Deployment cost	Embedded-friendly(268KB)	Low(1.2 MB)	Medium(1.5 MB)
Noise robustness	> 90% Recognition rate	~ 85%	~ 80%

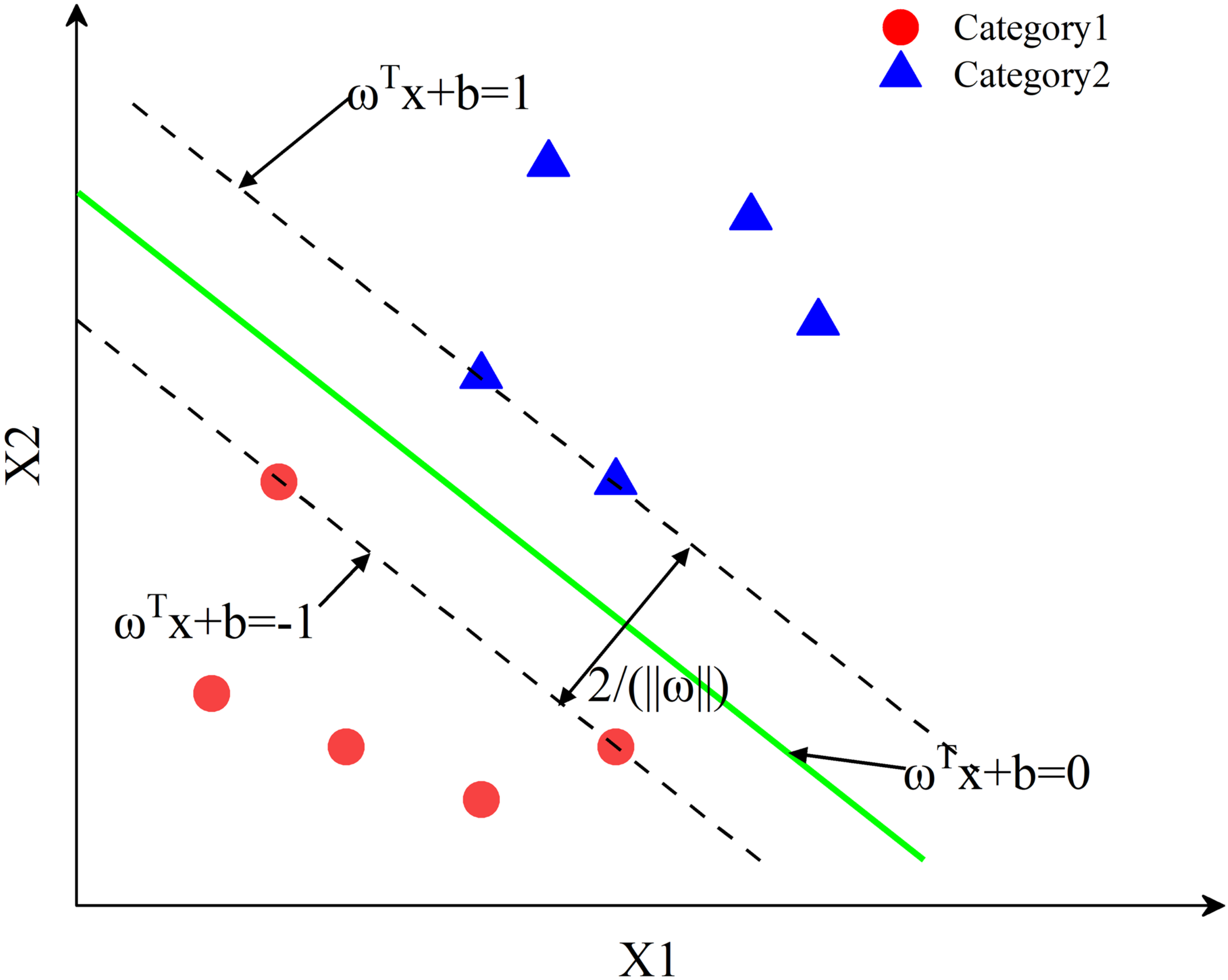


Fig. 1 SVM binary classification principle

Table 2 A brief introduction to the advantages and disadvantages of traditional machine learning models

Model	Advantages	Disadvantages	Interpretability	Overfit
Decision Tree	Mixed-data	Convergence-uncertain	Yes	Yes
K-NN	Low-code	Noise-vulnerable	Yes	Yes
Random Forest	High-dimension	Non-sequential	No	Yes
SVM	Overfit-resistant	Compute-heavy	Yes	No

3 Experiment and analysis

3.1 Detection system and detection steps

The architecture of the detection system primarily consists of several components, as shown in Fig. 2: a computer, industrial camera, optical lens, ring light illumination system, and testing platform. This detection system is implemented at the final quality control stage following manufacturing completion, conducting batch-based quality inspections on finished flange products. The detection process is as follows: First, the industrial camera captures an image of the flange end face. Next, the camera transmits the acquired image data to the computer system via a USB interface. Finally, on the computer, image processing is performed using PyCharm, which includes steps such as filtering and denoising, image segmentation, and edge detection. The detailed parameters of the detection system hardware configuration are provided in Table 3.

3.2 Algorithm flowchart

This paper proposes an improved Canny algorithm for edge detection of flange defects, combined with

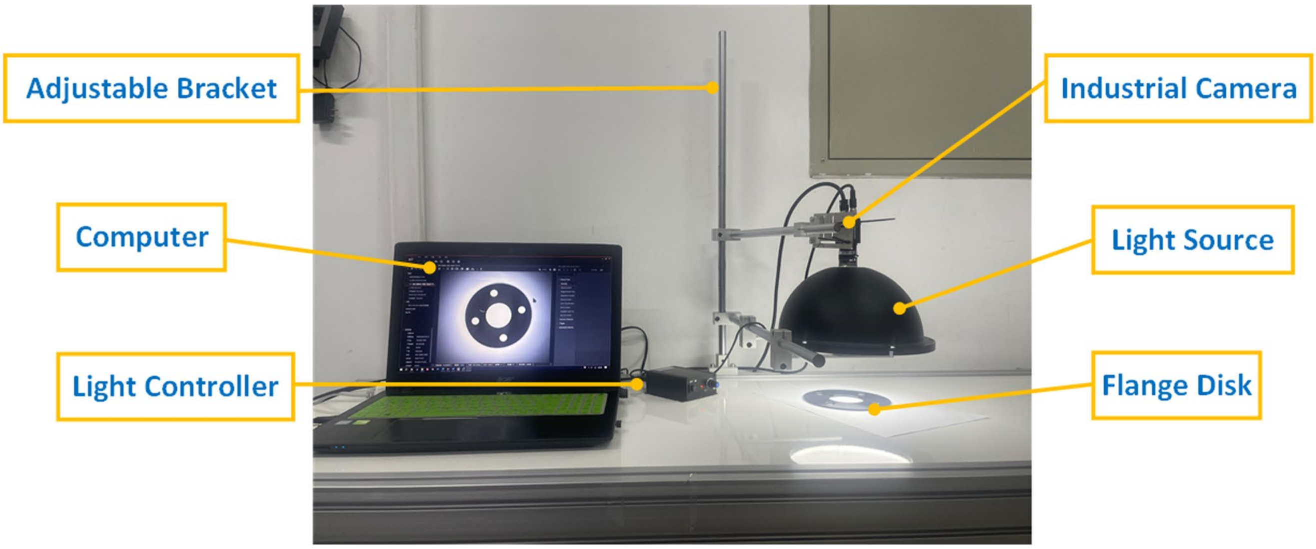


Fig. 2 Detection system platform

Table 3 Detect system hardware parameters

Hardware composition	Specifications
Industrial camera	MV-CS050-10GC
Lens	MVL-MF0824M-5MPE
Resolution	2448 × 2048
Light source	DBS-WD240
Flange disks	140 mm
Computer	Intel(R) Core(TM) i5-7200U

enhanced Hu moment feature extraction to train a Support Vector Machine (SVM) for defect classification. The specific process is as follows: First, image preprocessing operations are performed; then, the modified Canny algorithm is applied to the part images for preliminary localization to identify edge points. By connecting the edge points, the complete image contour is reconstructed. Next, enhanced Hu moment features are extracted from 600 sample images (representing three types of defects). Finally, a polynomial kernel function is chosen for training to achieve defect classification. The method steps are illustrated in Fig. 3.

3.3 Image acquisition and preprocessing

3.3.1 Image acquisition

This paper focuses on surface defect images of automotive flange disks. A total of 50 images were collected for each category, including images with sand holes, images with scratches, and defect-free images. The dataset was then augmented to 200 images per category using techniques such as flipping, translation, and adjustments to contrast and brightness. In the experiment, 70% of the total 600 images were randomly selected for the training set, while the remaining 30% were reserved for the test set, with 420 images in the training set and 180 images in the test set. Additionally, five images from each defect type were selected for display, as shown in Fig. 4.

3.3.2 Filtering and denoising

When capturing images of the flange disk, noise is often introduced. Five different denoising methods were used in this paper: mean filtering, Gaussian filtering, median filtering, bilateral filtering and improved bilateral filtering, and their denoising effects were compared. 100 randomly selected flange images were used to experimentally verify the denoising performance, with Peak Signal-to-Noise Ratio (PSNR) as the criterion for evaluating the denoising performance. The higher the PSNR value, the better the denoising effect. The experimental results are shown in Fig. 5. From the PSNR scatter plots of the different filters, it can be seen that the improved bilateral filtering algorithm demonstrated superior denoising performance, with a PSNR increase of 6.7% compared to the bilateral filter (which had the next best performance), a 9.1% increase compared to the median filter, 12.4% compared to the Gaussian filter, and 21.4% compared to the mean filter. Additionally, a further comparison of the denoising results of the five methods was conducted from a visual perspective, as shown in Fig. 6.

3.4 Image localization and segmentation

Due to factors such as uneven lighting, surface textures of the flange disk, and dust in the environment, the segmented image often contains not only defect features but

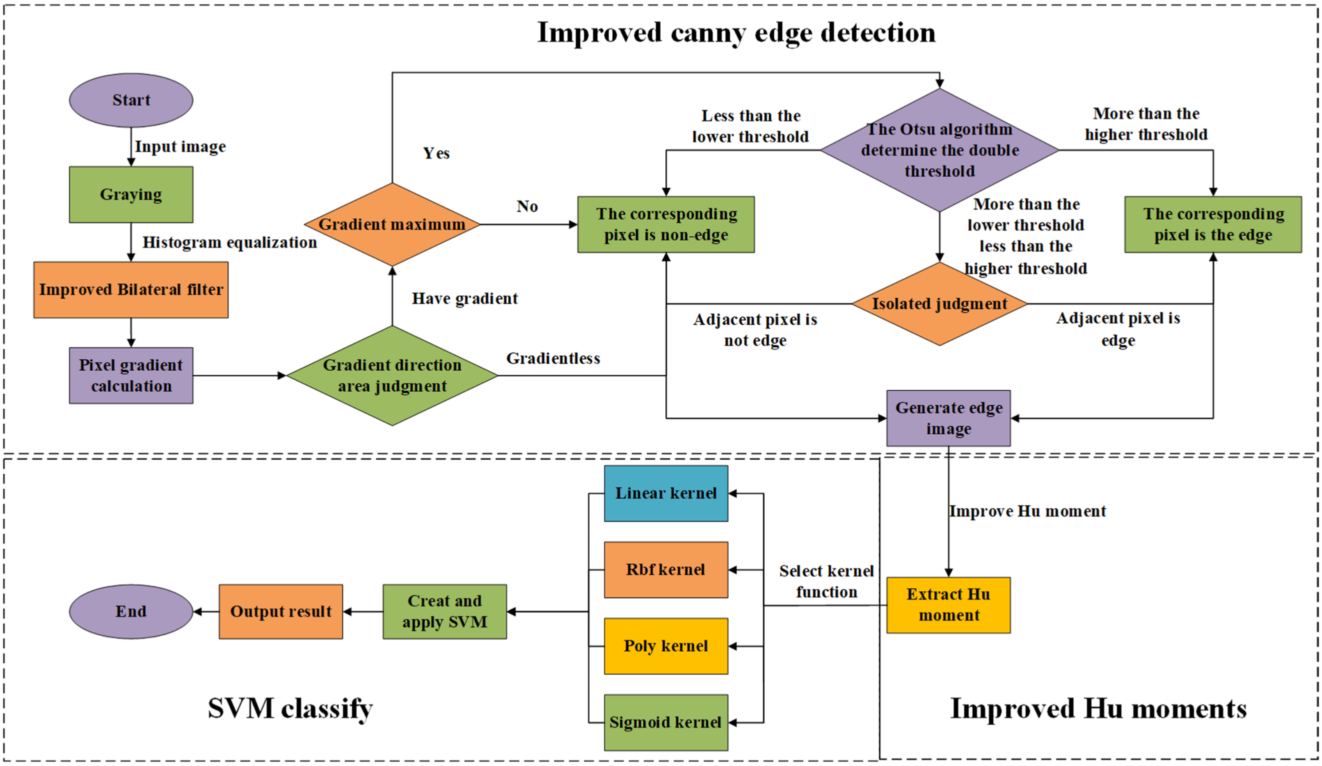


Fig. 3 Algorithm flowchart

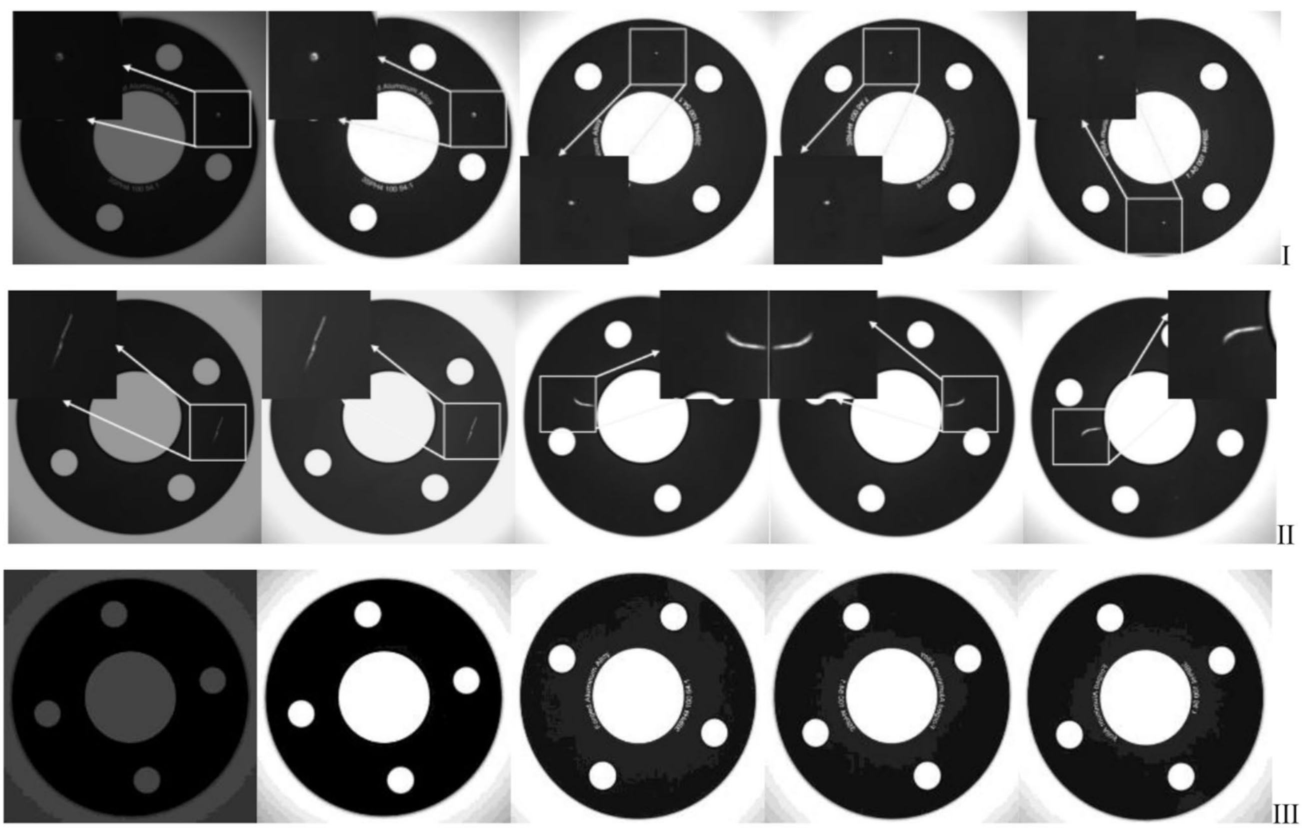


Fig. 4 Acquired raw images. **I** Sand hole defect **II** Scratch defect **III** No defect

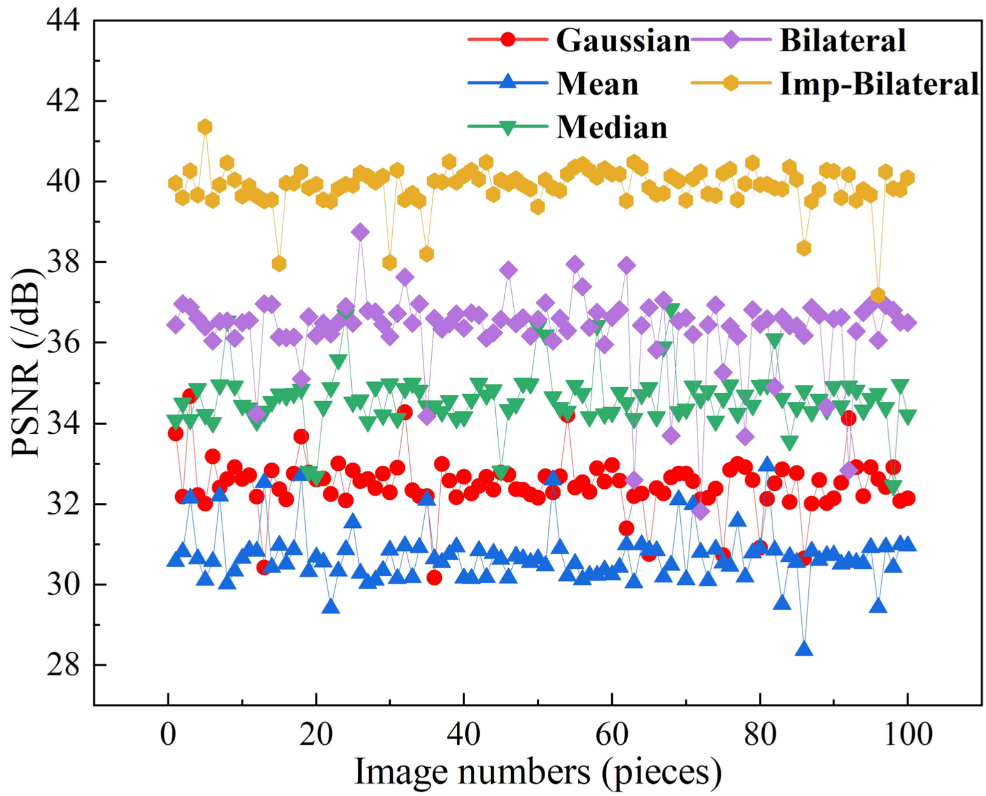


Fig. 5 Comparison of PSNR values for five filtering methods

also other disturbances, which hinder the precise extraction of defects. Therefore, it is necessary to perform interference removal, primarily by optimizing the foreground and background segmentation algorithms. The maximum entropy segmentation and Otsu method are two commonly used approaches [24]. After comparison, it was found that the Otsu method demonstrated superior performance in defect extraction. The specific results are shown in Fig. 7, where it effectively extracted defect features while significantly suppressing surrounding texture interference.

3.5 Experimental results analysis

3.5.1 Edge detection comparison

To evaluate the edge extraction performance of the improved Canny algorithm, several common detection methods [25] were compared, including the Sobel algorithm, standard Canny algorithm, and Laplacian algorithm, as shown in Fig. 8. From the observations, the following conclusions can be drawn: Fig. 8a shows that the Sobel algorithm exhibits discontinuous edges and misses some key information points. Figure 8b shows that the Laplacian algorithm detects severe shadows, and there is a noticeable color difference at the edges. Figure 8c demonstrates that the standard Canny algorithm produces a blurred result when extracting the flange disk contour, with significant noise in the circular hole region. In contrast, the improved Canny algorithm shows higher precision in extracting edge details and highlighting defect areas.

3.5.2 Kernel selection and ablation experiment

Before training the SVM, the enhanced Hu moments shape features are extracted as feature vectors, with partial extraction results shown in Table 4. The evaluation metrics for assessing the model's classification efficiency are accuracy, recall, and F1-score [26]. In determining the classification performance of different SVM kernel functions, this paper compared the performance of four kernels. As shown in Fig. 9, statistical analysis of the classification results reveals that the polynomial kernel exhibits outstanding performance across all evaluation metrics: it achieves an accuracy of 83.7%, a recall rate of 81.2%, and an F1 score of 82.4%. Compared to the linear kernel, radial basis function (RBF) kernel, and Sigmoid kernel, the polynomial kernel demonstrates a significant advantage in classification efficiency. Based on this, subsequent experiments will consistently use the polynomial kernel as the default kernel for the SVM model to enhance its prediction and classification capabilities.

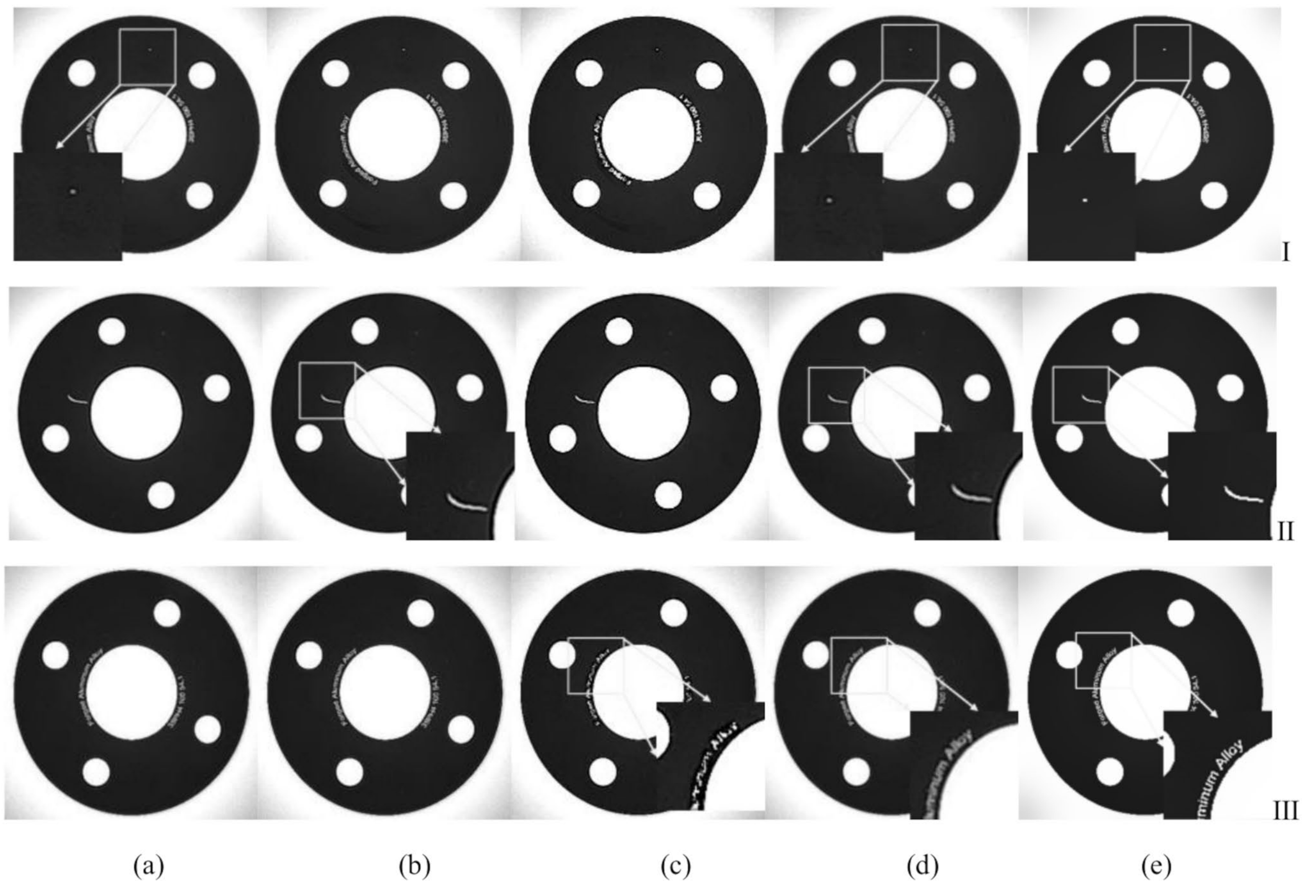
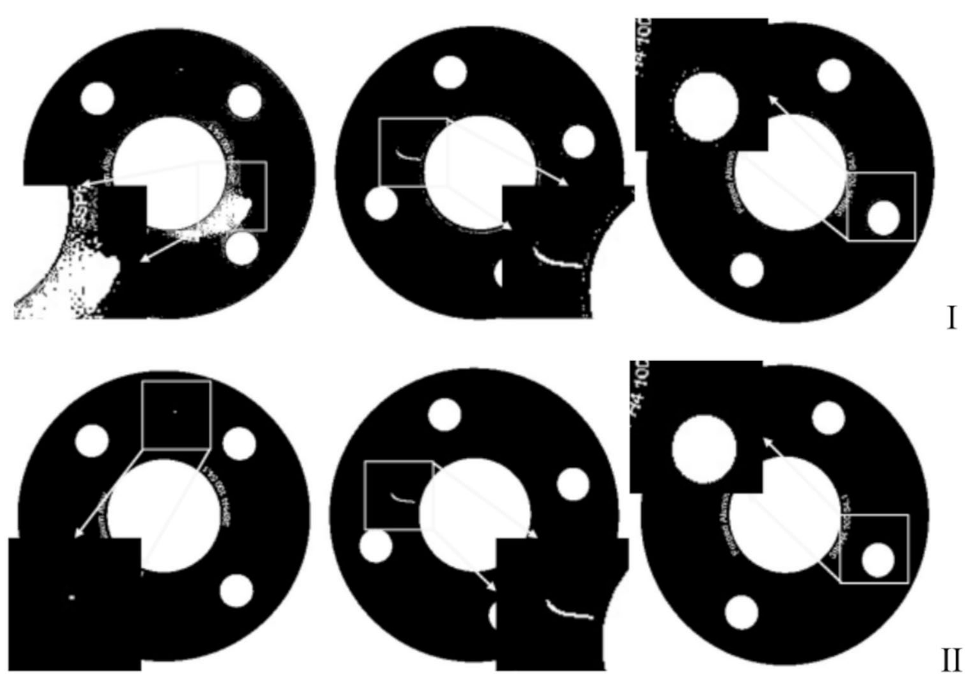


Fig. 6 Comparison of filtering effects. (**a**) Mean Filtering (**b**) Gaussian Filtering (**c**) Median Filtering (**d**) Bilateral Filtering (**e**) Improved Bilateral Filtering. **I** Sand hole defect **II** Scratch defect **III** No defect

Fig. 7 Comparison of image segmentation. **I** Maximum Entropy **II** Otsu Threshold



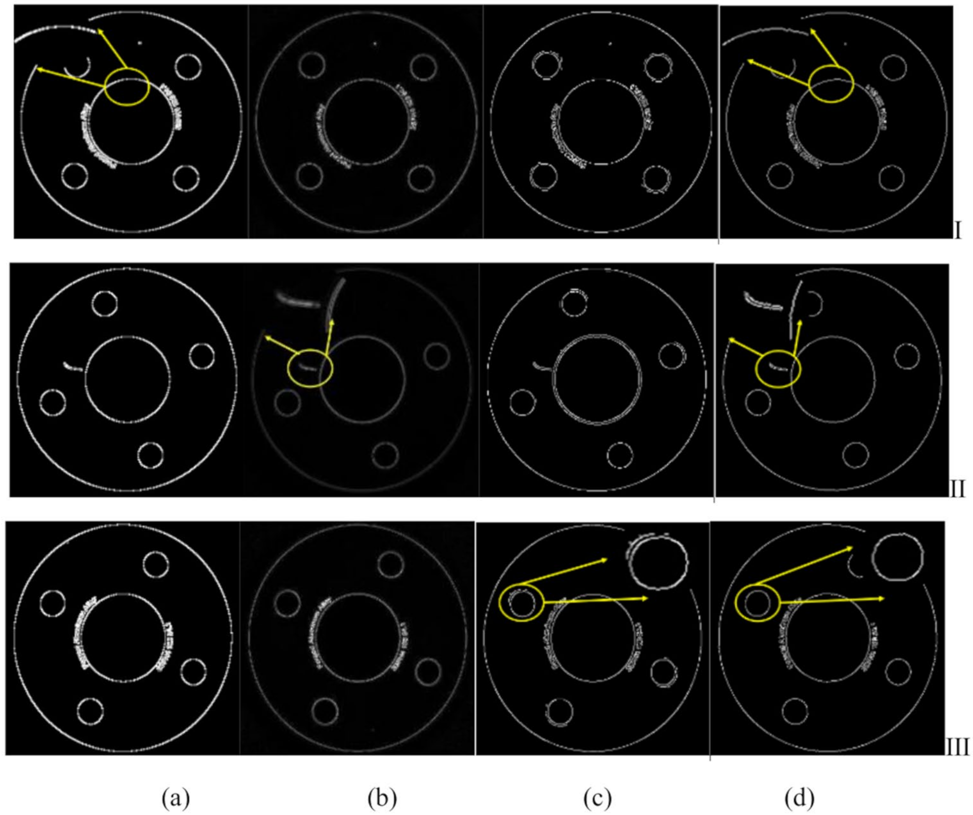


Fig. 8 Comparison of edge detection algorithms. (**a**) Sobel (**b**) Laplacian (**c**) Standard Canny (**d**) Improved Canny. **I** Sand hole defect **II** Scratch defect **III** No defect

Table 4 Feature extraction results

Image	ρ1	ρ2	ρ3	ρ4	ρ5	ρ6
Sand hole	0.238345	0.180647	0.163758	0.033364	0.205315	-0.035121
Scratch	0.311021	0.250114	0.183548	-0.050791	0.238743	0.053704
No defect	0.269423	0.219199	0.261355	0.048623	-0.266984	-0.048017

To rigorously validate the effectiveness of the proposed method, a series of ablation experiments were designed. Testing was conducted on 600 sample images to ensure the feasibility of the comparative results, with specific results shown in Table 5. original algorithm: In the Or-CannySVM method, the image is first subjected to traditional Canny edge detection, and then SVM training is performed based on the extracted edge information. The Or-HuSVM method directly extracts the original seven-dimensional Hu moment shape features from the image and uses them for SVM training. Improved algorithm: In the ImpCanny-SVM method, the image is first improved Canny edge detection, and then SVM training is performed based on the extracted edge information. The EnhHu-SVM method extracts the lightweight six-dimensional Hu moment shape features from the sample images and uses them for SVM training. Co-optimization: The proposed CannyHu-SVM model integrates these two improvement strategies, combining improved Canny edge detection with enhanced Hu moment shape features. As shown in Table 5, these improvements significantly improve the classification accuracy, fully demonstrating the effectiveness of the proposed enhancement strategy.

3.5.3 Comparison experiment

To verify the generalization performance of the model, this paper conducts a stability assessment of the CannyHu-SVM model using 5-fold cross-validation. The 600 images are randomly divided into 5 mutually exclusive subsets. In each iteration, 4 subsets (480 images) are used for training, and the remaining 1 subset (120 images) is used for testing. Finally, the average value of the five experiments is taken. As shown in Table 6, the results of the cross-validation indicate that the accuracy is 97.2% ± 0.6%, the recall rate is 96.3% ± 0.5%, and the F1-score is 96.8% ± 0.5%. These results are highly consistent with the results obtained from the original 70%/30% split, proving that the model has stable generalization ability.

In the dataset of 600 images (representing three types), labels 0, 1, and 2 are used to distinguish the image types for training purposes. Specifically, sand hole defects are

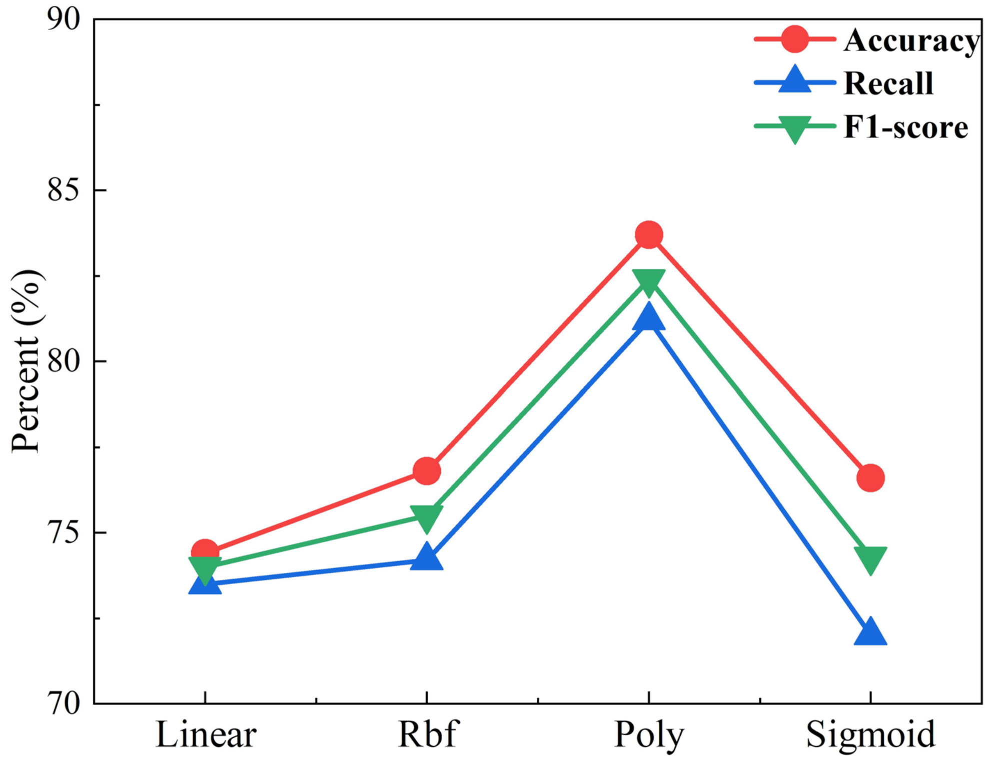


Fig. 9 Kernel performance comparison

Table 5 Ablation experiment

Model	Or-Canny	Or-Hu	Imp-Canny	Enh-Hu	Accuracy(%)	Recall(%)	F1-scroe(%)
SVM					83.7	81.2	82.4
OrCanny-SVM	√				88.6	86.4	87.0
OrHu-SVM		√			85.0	84.4	84.7
ImpCanny-SVM			√		90.0	88.2	89.1
EnhHu-SVM				√	86.1	85.6	85.9
CannyHu-SVM			√	√	**97.2**	**96.3**	**96.8**

Table 6 Five-fold cross-validation

Five-fold	Accuracy(%)	Recall(%)	F1-score(%)
1	97.0	96.1	96.6
2	96.7	95.8	96.2
3	97.5	96.6	97.1
4	**97.8**	**96.9**	**97.3**
5	97.2	96.3	96.8
Average	97.2	96.3	96.8
Standard deviation	±0.6%	±0.5%	±0.5%

labeled as 0, scratch defects are labeled as 1, and defect-free images are labeled as 2. To compare the high classification efficiency of the proposed model, prediction models were built using SVM, GA-SVM [27], WOA-SVM [28], and CannyHu-SVM, compared synchronous models with deep learning models. This paper selects lightweight networks MobileNetV3 [29] and EfficientNetB0 [30] as benchmarks, and output and analyze the prediction results of six models on the test set. The GA-SVM model primarily uses a genetic algorithm, which simulates natural selection and genetic mechanisms to find the optimal combination of penalty factor and kernel function parameters. The WOA-SVM model uses the hunting behavior of a group of humpback whales in nature to find the optimal solutions. The optimal values of c and r obtained by the two methods are 59.7836, 0.4295 for GA-SVM and 61.5627, 1.7814 for WOA-SVM. When training MobileNetV3 and EfficientNetB0, we input the edge processing results of the previous implementation and uniformly resized them to 224 × 224 resolution. The first convolutional layer was modified to accept 1-channel input instead of the original 3-channel RGB input, and the standard RGB normalization was replaced with (0,1) range scaling to accommodate the binary features from Canny edge detection. The models are initialized with ImageNet pretrained weights, with all layers unfrozen for end-to-end fine-tuning. The training protocol employs a learning rate of 0.0005, extends over 100 epochs to ensure sufficient convergence, utilizes the Adam optimizer, and sets the batch size to 8. In contrast, the proposed CannyHu-SVM model utilizes three parameters of the polynomial kernel to adjust the model. The specific steps are as follows: the scaling factor *r* is set to 0.5 to enhance the robustness of SVM. The independent coefficient *c* is set to 1. For the polynomial degree *d*, a degree of 3 is chosen to balance model complexity—avoiding underfitting caused by too low a degree and overfitting due to excessive complexity. The penalty factor *c* is set to 76.8429, and the model's *tol* for training is set to 0.001 to ensure that the algorithm stops iterating once a certain level of accuracy is achieved.

As shown in Fig. 10, the CannyHu-SVM and EfficientNetB0 models reduced misclassifications to 5, compared to 29 in the original SVM, 11 in GA-SVM, 9 in WOA-SVM, and 6 in MobileNetV3. Additionally, further comparisons were made regarding the prediction time of the models. A total of five experiments were conducted, and the results are shown in Fig. 11. As shown in Fig. 11, Compared with the GA-SVM and WOA-SVM models, the CannyHu-SVM demonstrates reduced average prediction times of 0.094s and 0.042s, respectively. When compared with the MobileNetV3 and EfficientNetB0 models, the average prediction time is further decreased by 2.956s and 5.556s. Compared to the fastest baseline SVM model, the CannyHu-SVM only increases the time cost by 0.279 s while improving accuracy by 13.5%. As shown in Table 7, the baseline SVM model achieves an average accuracy of merely 83.7%, indicating a higher error rate. In contrast, the GA-SVM, WOA-SVM and MobileNetV3 models achieve moderate accuracies of 93.8%, 95.0% and 96.5%, respectively. Although the accuracy of the EfficientNetB0 model method is slightly higher than that of CannyHu-SVM, the average time required is 9.1x that of the proposed method. In summary, the proposed CannyHu-SVM model achieves the accuracy of 97.2% and takes only 0.684s on average, demonstrating that this method provides more accurate, lower-error, and real-time performance for flange disk defect detection and classification. In practical applications, it can effectively reduce the occurrence of misclassification events.

3.6 Limitations and future directions

The model and method proposed in this paper still have certain limitations. For instance, the classification and recognition performance for flange disks may degrade under poor lighting conditions. Additionally, in terms of runtime and computational complexity, there is still room for improvement compared to the basic SVM model. Future work should focus on developing solutions to address these issues: (1) Actively explore adaptive illumination correction and background subtraction technologies, and develop a illumination invariance feature extraction module; (2) The multi-modal feature fusion mechanism is explored, and the two-dimensional features

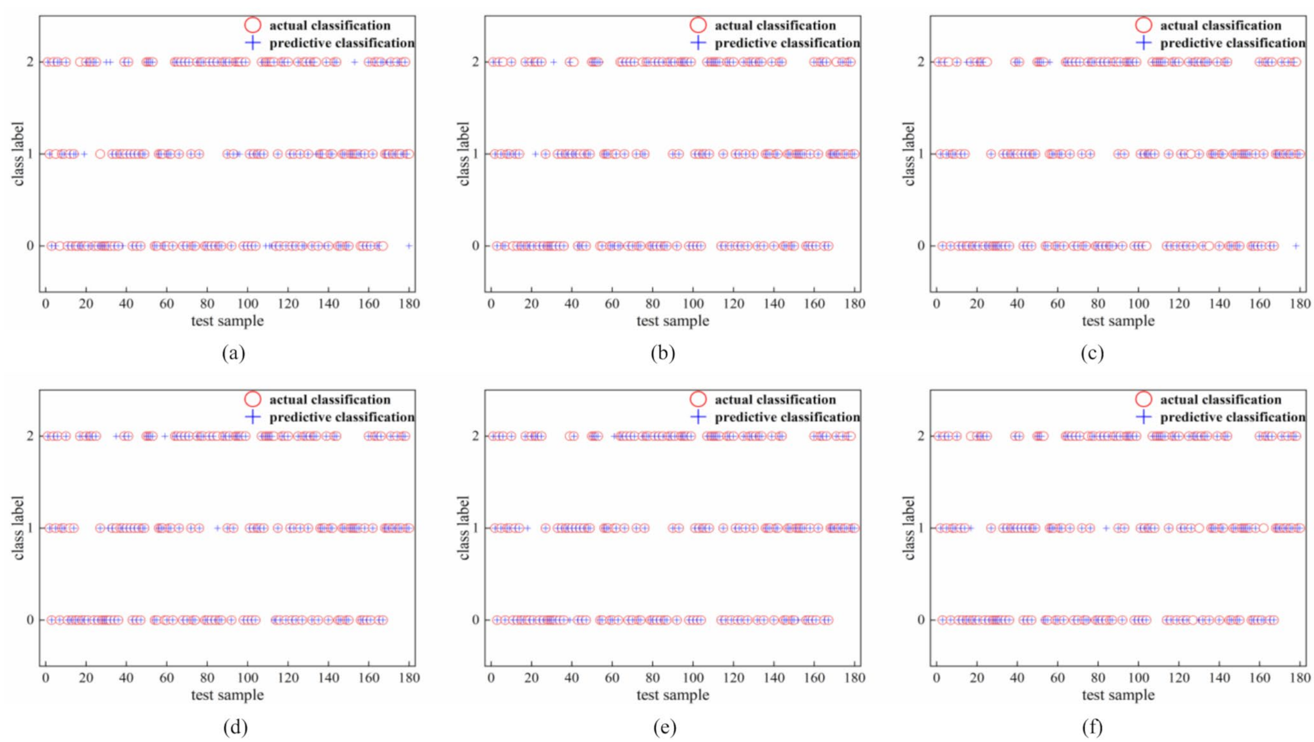


Fig. 10 Comparison of classification results. (**a**) SVM (**b**) GA-SVM (**c**) WOA-SVM (**d**) MobileNetV3 (**e**) EfficientNetB0 (**f**) CannyHu-SVM

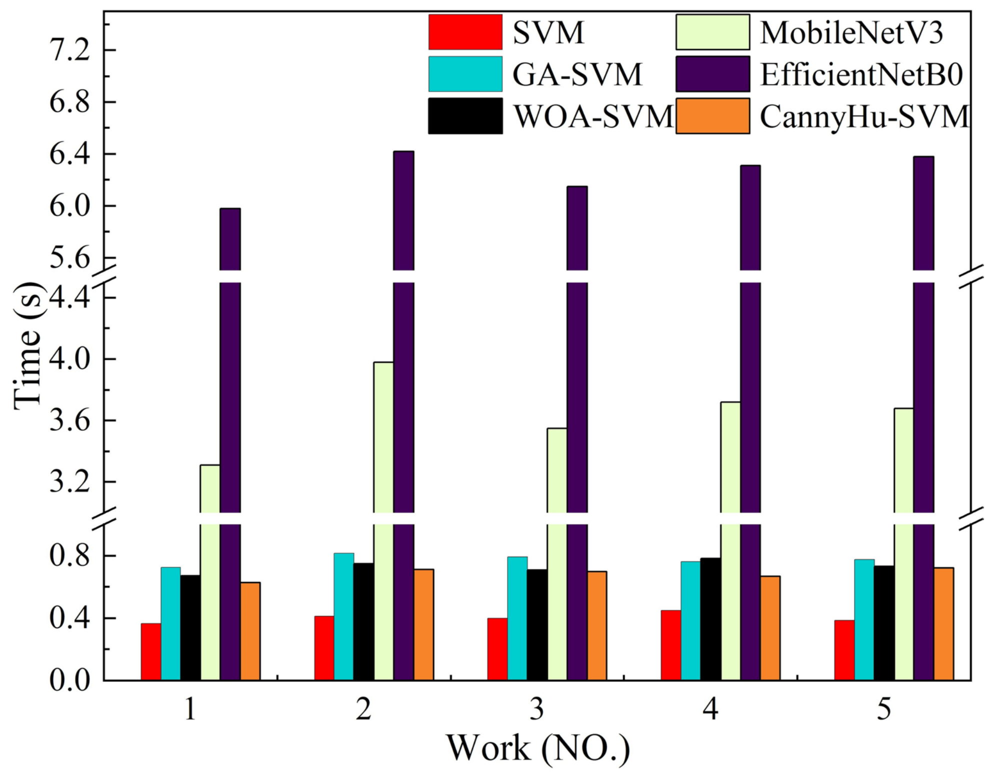


Fig. 11 Algorithm time comparison

Table 7 Comparison of results for six algorithms

Model	Accuracy (%)	Recall (%)	F1-score (%)	Average time spent (s)	Model Size (MB)
SVM	83.7	81.2	82.4	**0.405**	**0.15**
GA-SVM	93.8	91.6	92.7	0.778	0.3
WOA-SVM	95.0	93.8	94.4	0.726	0.43
MobileNetV3	96.5	94.5	95.5	3.64	3.8
EfficientNetB0	**97.4**	96.1	96.8	6.24	15.2
CannyHu-SVM	97.2	**96.3**	**96.8**	0.684	0.5

of the depth feature and the enhanced Hu moment are combined. (3) Research on FPGA-based hardware acceleration architecture and computer parallel processing technology; (4) Develop an online detection system, dynamically update the SVM kernel function parameters by using the real-time data of the production line, and establish a defect feature drift compensation mechanism to enhance the robustness and real-time performance of the model. (5) Deploy enclosed acquisition systems to physically isolate the inspection environment, minimizing dust interference through controlled air filtration and sealing mechanisms explore backlight illumination schemes for edge detection to replace conventional positive light sources. By projecting uniform backlighting, the edge profiles of flange disks can be captured with minimal interference from surface textures and dust. The goal is to ensure efficient classification in practical applications while overcoming the challenges posed by external factors such as lighting conditions.

4 Conclusion

This paper presents a model, referred to as CannyHu-SVM, designed to address the defect classification problem in the flange disk production process within industrial automation. Building on the standard Canny algorithm, the method compares five different filtering techniques and evaluates them using PSNR values. Improved gradient bilateral filtering, which demonstrated superior denoising performance, is selected to replace Gaussian filtering for preprocessing the acquired flange disk images. This effectively suppresses the impact of noise on image analysis. Two image segmentation methods are compared, and the Otsu method is chosen for segmentation, as it suppresses texture interference while preserving details. Through the improvement of Hu moments, the method effectively overcomes the influence of scaling factors, maintaining scale-invariance while also ensuring rotational and translational invariance. Finally, the polynomial kernel was selected as the kernel for the SVM, achieving an average classification accuracy of 97.2% for the three types of flange disks. This performance surpasses that of the baseline SVM, GA-SVM, WOA-SVM and MobileNetV3 models, its prediction accuracy is close to that of the EfficientNetB0 model, while its average inference time is

9.1 times faster than that of the EfficientNetB0 model, with lower prediction time. This provides accurate and efficient theoretical support for flange disk defect detection research.

Author contributions C. Y. G proposed the CannyH-SVM improvement method and completed the experiments. J. G supervised the writing of the paper and helped to revise and embellish it. M. S. M and H. M. F completed the processing of experimental data. Q. Y. Z, S. Q. L and L. S. Y produced the Tables 1, 2, 3, 4, 5, 6 and 7; figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval Not applicable.

Consent for publication Author agreed with the content and that all gave explicit consent to submit.

Competing interests The authors declare no competing interests.

References

1. Singh, H., Weng, B., Rao, S.J., et al.: A diversity analysis of safety metrics comparing vehicle performance in the lead-vehicle interaction regime[J]. IEEE Trans. Intell. Transp. Syst. (2023). https://doi.org/10.1109/tits.2023.3290261
2. Kim, T.-K., Zafeiriou, S., Glocker, B., et al.: Special issue on machine vision[J]. Int. J. Comput. Vision. **27**, 1611–1613 (2019). https://doi.org/10.1007/s11263-019-01201-4
3. Gómez-Martínez, V., Chushig-Muzo, D., Veierød, M.B., et al.: Ensemble feature selection and tabular data augmentation with generative adversarial networks to enhance cutaneous melanoma identification and interpretability[J]. BioData Min. **17**(1), 46 (2024). https://doi.org/10.1186/s13040-024-00397-7
4. Rex, C.E.S., Annrose, J., Jose, J.J.: Comparative analysis of deep convolution neural network models on small scale datasets[J]. Optik. **271**, 170238 (2022). https://doi.org/10.1016/j.ijleo.2022.170238
5. Voulodimos, A., Doulamis, N., Doulamis, A., et al.: Deep learning for computer vision: A brief review[J]. Comput. Intell. Neurosci. **2018**(1), 7068349 (2018). https://doi.org/10.1155/2018/7068349
6. Injadat, M., Moubayed, A., Nassif, A.B., et al.: Machine learning towards intelligent systems: applications, challenges, and opportunities[J]. Artif. Intell. Rev. **54**(5), 3299–3348 (2021). https://doi.org/10.1007/s10462-020-09948-w
7. Li, Y., He, T.: Identification and detection of apple leaf diseases based on GA-SVM and feature extraction[J]. Food Ferment. Technol. **60**(03), 7–13 (2024). https://doi.org/10.3969/j.issn.1674-506X.2024.03-002
8. Liu, R., Mao, J., Lu, W.: Research on visual inspection methods for appearance defects of automobile precision parts[J]. Comput. Digit. Eng. **49**(02), 383–387 (2021). https://doi.org/10.3969/j.issn.1672-9722.2021.02.029
9. Zhu, J., Li, Y., Tang, Z.: Strip surface defect detection based on PSO-SVM[J]. Journal of Changchun University of Science and Technology (Natural Science Edition). **47**(01), 42–51 (2024). https://doi.org/10.1672/9870-01-0042-10
10. Zhang, Z., Yu, N., Bian, Y., et al.: Research on emotion recognition methods based on multimodal physiological signal feature fusion[J]. J. Biomed. Eng. 1–7 (2025). https://doi.org/10.7507/1001-5515.20-24010-20
11. Mao, Y., Fang, L., Ye, X., et al.: A rapid photoionization detector-based gas chromatography electronic nose enhanced by DOE and SVM for accurate classification of Baijiu Jiuqu quality grades[J]. J. Food Compos. Anal. **107302** (2025). https://doi.org/10.1016/j.jfca.2025.107302
12. Wu, M., Yan, C., Sen, G.: Computer-aided diagnosis of hepatic cystic echinococcosis based on deep transfer learning features from ultrasound images[J]. Sci. Rep. **15**(1), 607 (2025). https://doi.org/10.1038/s41598-024-85004-9
13. Tan, L.: Research on automatic detection of spark plug defects based on machine vision technology[J]. Small internal combustion engines and vehicle technology. **52**(05), 81–86 (2023). https://doi.org/10.2095/8234/05-0081-06
14. Liang, J., Yuan, X., Li, H., et al.: Research on surface defect detection method of lamp cover based on machine vision[J]. Mod. Industrial Econ. Informatization. **13**(12), 131–136 (2023). https://doi.org/10.16525/j.cnki.14-1362/n.2023.12.042
15. Xie, D., He, F., He, H., et al.: Machine vision-based surface defect detection of wipers[J]. Comput. Digit. Eng. **52**(04), 1221–1227 (2024). https://doi.org/10.3969/j.issn.1672-9722.2024.04.047
16. Qi, J., Zhang, J., Dai, X., et al.: Defect detection of parts based on machine vision[J]. Smart computers and applications. **11**(03), 167–171 (2021). https://doi.org/10.2095/2163/03-0167-05
17. Cai, B., Huang, D., Zhang, H., et al.: Online detection method of bearing full-surface defects based on machine vision[J]. Electronic measurement technology. 1–11 (2025). https://doi.org/10.19651/j.cnki.emt.2417502
18. Malmir, S., Shalchian, M.: Design and FPGA implementation of dual-stage lane detection, based on Hough transform and localized stripe features[J]. Microprocess. Microsyst. **64**, 12–22 (2019). https://doi.org/10.1016/j.micpro.2018.10.003
19. Lu, Q., Han, Z., Hu, L., et al.: An infrared and visible image fusion algorithm method based on a dual bilateral least squares hybrid filter[J]. Electronics. **12**(10), 2292 (2023). https://doi.org/10.3390/electronics12102292
20. Rangarajan, A., Chellappa, R., Zhou, Y.: A model-based approach for filtering and edge detection in noisy images[J]. IEEE Trans. Circuits Syst. **37**(1), 140–144 (1990). https://doi.org/10.1109/31.45704
21. Wenwen, M., Zixin, H., Zha, L., et al.: Image-free Hu invariant moment measurement by single-pixel detection[J]. Opt. Laser Technol. **181**, 111581 (2025). https://doi.org/10.1016/j.optlastec.2024.111581
22. Prabhavathy, T., Elumalai, V.K., Balaji, E.: Gesture recognition framework for upper-limb prosthetics using entropy features from electromyographic signals and a Gaussian kernel SVM classifier[J]. Appl. Soft Comput. **167**, 11238 (2024). https://doi.org/10.1016/j.asoc.2024.112382
23. Jia, H., Guo, Y., Ma, Q., et al.: A review of spot electricity price forecasting based on machine learning methods[J]. Power Constr. **46**(02), 160–179 (2025). https://doi.org/10.12204/j.issn.1000-7229.2025.02.014
24. Garcia-Lamont, F., Cervantes, J., López, A., et al.: Segmentation of images by color features: A survey[J]. Neurocomputing. **292**, 1–27 (2018). https://doi.org/10.1016/j.neucom.2018.01.091
25. Tariq, N., Hamzah, R.A., Ng, T.F., et al.: Quality assessment methods to evaluate the performance of edge detection algorithms for digital image: A systematic literature review[J]. IEEE

Access. **9**, 87763–87776 (2021). https://doi.org/10.1109/ACCESS.2021.3089210
26. Sharma, S., Timilsina, S., Gautam, B.P., et al.: Enhancing sika deer identification: integrating CNN-based Siamese networks with SVM classification[J]. Electronics. **13**(11), 2067 (2024). https://doi.org/10.3390/electronics13112067
27. Liu, X., Han, Q., Zhou, Y., et al.: Research on the application of liquor classification and identification method based on GA optimization SVM parameters[J]. Packaging Food Mach. **40**(02), 64–68 (2022). https://doi.org/10.3969/j.issn.1005-1295.2022.02.012
28. He, X., Zhang, G., Chen, Y., et al.: WOA-SVM multi-classification algorithm integrating Lévy flight and elite reverse learning[J]. Comput. Appl. Res. **38**(12), 3640–3645 (2021). https://doi.org/10.19734/j.issn.1001-3695.2021.04.0164
29. Heng, Y., Sheng, Z., Yan, Y., et al.: Sound classification and recognition method of caged laying hens based on improved MobileNetV3[J]. Trans. CSAM. **56**(04), 427–435 (2025). https://doi.org/10.6041/j.issn.1000-1298.2025.04.040
30. Assaduzzaman, M., Bishshash, P., Nirob, M. a: S, XSE-TomatoNet: An explainable AI based tomato leaf disease classification method using EfficientNetB0 with squeeze-and-excitation blocks and multi-scale feature fusion[J], **14**: 103159 (2025). https://doi.org/10.1016/j.mex.2025.103159